

for Hardy-Weinberg equilibrium states that if a population is not evolving at a locus with two alleles, the three genotypes will appear in the following proportions:

$$p^2 + 2pq + q^2 = 1$$

Expected frequency of genotype  $C^RC^R$

Expected frequency of genotype  $C^RC^W$

Expected frequency of genotype  $C^WC^W$

Note that for a locus with two alleles, only three genotypes are possible (in this case,  $C^RC^R$ ,  $C^RC^W$ , and  $C^WC^W$ ). As a result, the sum of the frequencies of the three genotypes must equal 1 (100%) in *any* population regardless of whether the population is in Hardy-Weinberg equilibrium. The key point is that a population is in Hardy-Weinberg equilibrium only if the observed genotype frequency of one homozygote is  $p^2$ , the observed frequency of the other homozygote is  $q^2$ , and the observed frequency of heterozygotes is  $2pq$ . Finally, as suggested by Figure 23.8, if a population such as our wildflowers is in Hardy-Weinberg equilibrium and its members continue to mate randomly generation after generation, allele and genotype frequencies will remain constant. The system operates somewhat like a deck of cards: No matter how many times the deck is reshuffled to deal out new hands, the deck itself remains the same. Aces do not grow more numerous than jacks. And the repeated shuffling of a population’s gene pool over the generations cannot, in itself, change the frequency of one allele relative to another.

Conditions for Hardy-Weinberg Equilibrium

The Hardy-Weinberg approach describes a population that is not evolving. This can occur if a population meets all five of the conditions for Hardy-Weinberg equilibrium listed in Table 23.1. But in nature, the allele and genotype frequencies of a population often *do* change over time. Such changes

| Table 23.1 Conditions for Hardy-Weinberg Equilibrium |                                                                                                                                                                                         |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Condition                                            | Consequence if Condition Does Not Hold                                                                                                                                                  |
| 1. No mutations                                      | The gene pool is modified if mutations occur or if entire genes are deleted or duplicated.                                                                                              |
| 2. Random mating                                     | If individuals mate within a subset of the population, such as near neighbors or close relatives (inbreeding), random mixing of gametes does not occur and genotype frequencies change. |
| 3. No natural selection                              | Allele frequencies change when individuals with different genotypes show consistent differences in their survival or reproductive success.                                              |
| 4. Extremely large population size                   | In small populations, allele frequencies fluctuate by chance over time (genetic drift).                                                                                                 |
| 5. No gene flow                                      | By moving alleles into or out of populations, gene flow can alter allele frequencies.                                                                                                   |

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can occur when at least one of the conditions for Hardy-Weinberg equilibrium is not met. Although departure from the conditions in Table 23.1 is common—resulting in evolutionary change—it is also common for natural populations to be in Hardy-Weinberg equilibrium for specific genes. One way this can happen is if selection alters allele frequencies at some loci but not others. In addition, some populations evolve so slowly that the changes in their allele and genotype frequencies are difficult to distinguish from those predicted for a non-evolving population.

Applying the Hardy-Weinberg Equation

The Hardy-Weinberg equation is often used as an initial test of whether evolution is occurring in a population (Concept Check 23.2, question 3 is an example). The equation also has medical applications, such as estimating the percentage of a population carrying the allele for an inherited disease. For example, consider phenylketonuria (PKU), a metabolic disorder that results from homozygosity for a recessive allele. This disorder occurs in about one out of every 10,000 babies born in the United States. Left untreated, PKU results in mental disability and other problems. (As described in Concept 14.4, newborns are now routinely tested for PKU, and symptoms can be largely avoided with a diet very low in phenylalanine.)

To apply the Hardy-Weinberg equation, we must assume that no new PKU mutations are being introduced into the population (condition 1) and that people neither choose their mates on the basis of whether or not they carry this gene nor generally mate with close relatives (condition 2). We must also ignore any effects of differential survival and reproductive success among PKU genotypes (condition 3) and assume that there are no effects of genetic drift (condition 4) or of gene flow from other populations into the United States (condition 5). These assumptions are reasonable: The mutation rate for the PKU gene is low, inbreeding and other forms of nonrandom mating are not common in the United States, selection occurs only against the rare homozygotes (and then only if dietary restrictions are not followed), the U.S. population is very large, and populations outside the country have PKU allele frequencies similar to those seen in the United States.

If all these assumptions hold, then the frequency of individuals in the population born with PKU will correspond to  $q^2$  in the Hardy-Weinberg equation ( $q^2$  = frequency of homozygotes). Because the allele is recessive, we must estimate the number of heterozygotes rather than counting them directly as we did with the pink flowers. Recall that there is one PKU occurrence per 10,000 births, which indicates that  $q^2$  = 0.0001. Thus, the frequency ( $q$ ) of the recessive allele for PKU is

$$q = \sqrt{0.0001} = 0.01$$

and the frequency of the dominant allele is

$$p = 1 - q = 1 - 0.01 = 0.99$$